MILITARY SPECIFICATION

CARTRIDGE, ILLUMINATING, M301A2 WITH FUZE, TIME, M84 FOR 81MM MORTARS LOADING, ASSEMBLING AND PACKING

This amendment forms a part of Military Specification MIL-C-46919A (MU), dated 15 January 1965 and is approved for use by the US Army Armament Research and Development Command and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

Add the following preamble underneath the title:

"This specification is approved for use by the US Army Armament Research and Development Command and is available for use by all Departments and Agencies of the Department of Defense."

2.1, Drawings: Delete "7548648 and 7548649" in their entirety and substitute the following:

"9241848 - Container, Ammunition Fiber, PA43, for Cartridge, 81MM, Illuminating, M301A2 and M301A3, in Jungle Wrap.

9241850 - Protective Bag Assembly for Cartridge, 81MM, Illuminating, M301A2 and M301A3, in Jungle Wrap.

9241849 - Box, Packing, Ammunition, for Cartridge, 81MM, Illuminating, M301A2 and M301A3, in Jungle Wrap."

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3.4.4, line 4: Delete "(see 6.6)" and substitute "(see 6.5)".

3.4.5, line 2: After "within" add "approximately".

FSC 1315

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PAGE 5

4.1.2: Delete in its entirety.

PAGE 6

4.2.2.2:

Critical defects 4 and 5: Delete in their entirety and substitute the following:

"4. More than two propellant increments missing...Visual 02004.

5. Fin broken or cracked ... Visual 02005."

Critical defect 6: Delete "or inverted."

Major defect 101: Delete in its entirety.

Major defect 105: Delete "(except as described by major defect 101 above)" and substitute "(except as described in any other defect contained herein)."

Add new Major defect 106 for 100 Percent Inspection as follows:

" 100 Percent Inspection

106. Any propellant increment missing ... Visual 02022."

Add new Major defect 107 as follows:

"107. Fin distorted ... Visual/Gage 02023."

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4.2.2.3: Delete in its entirety and substitute the following:

"4.2.2.3 Unsealed fiber container (see dwg. 9241848 and dwg. 9241850).

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Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 Percent		
101.	Neck ring missing	Visual	03001
102.	Support tube missing	Visual	03002
103.	Bag missing	Visual	03003
104.	Any filler material missing	Visual	03004
Minor:	AQL 1.5 Percent		
201.	Wire tie not properly secured	Visual	03005
202.	Desiccant missing	Visual	03006
203.	Round cannot be removed by hand	Manual	03007
204.	Evidence of poor workmanship (see 3.5)."	Visual	03008

4.2.2.4: Delete in its entirety and substitute the following:

"4.2.2.4 Sealed fiber container wrapped in barrier material prior to wax jungle wrap (see dwg. 9241848 and dwg. 9230175).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 percent		
101.	Sealing tape incomplete or badly wrinkled	Visual	04001
102.	Barrier material improperly assembled	Visual	04002
103.	Barrier material improperly folded or taped	Visual	04003
104.	Cartridge loose in container	Manual	04004

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Categories	Defects	Method of Inspection	Code No.
Minor:	AQL 2.5 percent		
201.	Gap between body and cover of container exceeding 1/8 inch	Visual/Gage	04005
202.	Tear tab length inadequate	Visual	04006
203.	Marking misleading or unidentifiable	Visual	04007
204.	Marking missing or incomplete	Visual	04008
205.	Evidence of poor workmanship (see 3.5)."	Visual	04009

4.2.2.5: Delete "dwg. 7458649" and substitute "dwg. 9241849."

Add new paragraph 4.2.2.6 as follows:

"4.2.2.6 Sealed fiber container after waxing - Jungle wrap (see dwg. 9241848).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.4 Percent		
101.	Wax coating incomplete or damaged	Visual	06001
Minor:	AQL 1.00 Percent		
201.	Marking missing or unidentifiable	Visual	06002
202.	Kraft paper improperly placed on container."	Visual	06003

4.2.3.1, last line: Delete "(see 6.7)" and substitute "(see 6.6)".

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4.2.3.2, line 2: Delete "02014" and substitute "02021."

Delete paragraph 4.2.3.3 Table I, 4.2.3.1.1, and 4.2.3.1.2 in their entirety and substitute the following:

"4.2.3.3 Functioning

4.2.3.3.1 Beginning with the first lot produced and continuing until three consecutive lots have complied with the acceptance criteria specified in Table I, eighty sample cartridges shall be selected for this test. The cartridges shall be observed for compliance with the requirements specified. Testing shall be in accordance with 4.3.2.

4.2.3.3.2 After three consecutive lots have complied with the criteria of 4.2.3.3.1, thirty-two sample cartridges shall be selected for this test. The cartridges shall be observed for compliance with Table I and the requirements specified. Testing shall be in accordance with 4.3.2.

TABLE I

Category	Defect	Code No.	Sample Size 80 (see 4.2.3.3.1) Acc.	Rej.*	Sample Size 80 (see 4.2.3.3.2) Acc.	Rej.*
Critical	Premature burst or metal parts separation	02014	0	1	0	1
Major	Failure to fire	02015	1	2	0	1
Major	Fuze Functioning	02016	4	5	2	3
Major	Horizontal range less than 80% of average range	02017	1	2	0	1
Major	Projectile does not ignite or provides for effective illumination for 40 sec. or less	02018	5	6	3	4

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Category	Defect	Code No.	Sample Size 80 (see 4.2.3.3.1) Acc.	Rej.*	Sample Size 80 (see 4.2.3.3.2) Acc.	Rej.*
Minor	Projectile provides effective illumination for more than 40 seconds but less than 60 seconds	02019	8	9	5	6

* NOTE: The lot shall also be rejected if in a sample of 80 rounds, a total of eight or more major defects is encountered.

** NOTE: The lot shall also be rejected if in a sample of 32 rounds, a total of five or more defects is encountered."

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4.2.4: Delete in its entirety.

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4.3.2: Delete in its entirety and substitute the following:

"4.3.2 Proving Ground. The test shall be performed at a Government Proving Ground in accordance with the Applicable Test Procedure. All of the cartridges shall be fired from an 81MM Mortar for which the round is standard and four propellant increments shall be used on each cartridge. The cartridges shall be fired at an elevation of 1060 mils with the fuze set at 23.5 seconds. However, if meterological conditions or time of fuze functioning so dictate, the proving ground shall adjust the fuze in order to increase the height of burst and insure effective illumination (see 6.5). Observation shall be made for compliance with the requirements specified and in addition, fuze functioning time recorded to the nearest 0.1 second, illuminant burning time to the nearest 0.5 second and horizontal range on the ground to a point directly under the position of fuze

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functioning to the nearest 5 meters measured by optical instrument triangulation. Any malfunction occurring during the test shall be reported in the firing record. Meterological data shall also be reported in the official firing record."

4.3.3: Delete in its entirety.

Delete section 5 in its entirety and substitute the following:

"5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Level A. The cartridges shall be packaged in accordance with dwg. 9241848 and dwg. 9241850.

5.2 Packing

5.2.1 Level A. The packaged cartridges shall be packed in accordance with dwg. 9241849.

5.3 Marking

5.3.1 The marking of the packages and pack shall be in accordance with dwg. 9241848 and dwg. 9241849.

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6.1: Delete in its entirety and substitute the following:

"6.1 Ordering data. Procurement document shall specify the following:

a. Title, number and date of this specification.

b. Data cards shall be prepared for each lot in accordance with the information specified in MIL-STD-1167."

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6.4: Delete in its entirety.

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6.5: Delete in its entirety and substitute the following:

"6.5 Effective illumination. Effective illumination is defined as burning of the canister without exploding or breaking up while suspended by the parachute. The time recorded for "effective illumination" shall be considered as the time from the appearance of first light from the candle until either: (1) burnout of the candle while still in the air (normal burnout) or (2) ground impact. Burning of the canister after striking the ground shall not be considered as "effective illumination". Any occurrences which detract from "effective illumination" as herein defined, shall be recorded on the official firing record."

Add new paragraph 6.7 as follows:

"6.7 Submission of test data. In addition to the normal distribution of records, when the cartridge is procured by the Department of the Army, one copy of all ballistic test data shall be forwarded to the following: Commander, USA ARRADCOM, Dover, N.J. 07801, ATTN: QAT-A.

Custodian:
Army-AR

Preparing Activity:
Army-AR

(Project 1315-A463)

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